

Fifth Assignment

14.11.2024

Sent

14/14 Points

Forsøk 3



Se gjennom tilbakemelding

15.11.2024

Forsøk 3 Poengsum:

14/14



Vis tilbakemelding

Anonym vurdering: **nei**

3 tillatte forsøk

5.11.2024 til 15.11.2024

▼ Detaljer

Objective

Your task is to design and implement a spell-checker class in C++ that checks the spelling of a given word based on a predefined set of words. This class should be efficient in storing and retrieving words using a hash table.

Description:

Write a spell-checker class that stores a set of words, W , in a hash table and implements a function, `spellCheck(s)`, which performs a spell check on the string s with respect to the set of words, W . If s is in W , then the call to `spellCheck(s)` returns an iterable collection that contains only s , since it is assumed to be spelled correctly in this case. Otherwise, if s is not in W , then the call to `spellCheck(s)` returns a list of every word in W that could be a correct spelling of s . Your program should be able to handle all the common ways that s might be a misspelling of a word in W , including swapping adjacent characters in a word, inserting a single character in between two adjacent characters in a word, deleting a single character from a word, and replacing a character in a word with another character. For an extra challenge, consider phonetic substitutions as well. (**Reference book, Page 420, P-9.1**).

Requirements

Class Design (2 points)

Define a class `SpellChecker` with the following:

- A private hash table data structure to store a set of correctly spelled words, W .
- A constructor that takes a list of words and initializes the hash table with these words.
- A public function `spellCheck(std::string s)` that takes a string s and returns a list of potential correct spellings for s
- Choose a good hash function.

Spell Check Function (3 points)

- Implement the `spellCheck(std::string s)` function to check if the word `s` exists in the hash table `W`.
- If `s` is found in `W`, return a collection containing only `s`, as it is assumed to be spelled correctly.
- If `s` is not in `W`, `spellCheck(s)` should return a list of potential matches from `W`. This list should include every word in `W` that could be a correct spelling of `s` based on the following misspelling conditions:
 - Swapping: Swapping any two adjacent characters in `s`.
 - Insertion: Inserting a single character between any two adjacent characters in `s`.
 - Deletion: Deleting a single character from `s`.
 - Replacement: Replacing any character in `s` with another character.

Testing and User Interface (2 points)

Implement a user-friendly interface. Test your spell-checker on various inputs, ensuring that it accurately identifies correctly spelled words and provides a list of possible corrections for misspelled words.

Exception Handling (1 point)

- Implement exception handling.

Phonetic Substitutions and documentation (2 points)

Implement phonetic similarity checks for additional accuracy in misspelling detection. For example, allow common phonetic substitutions, such as "ph" for "f" or "c" for "k".

Please make sure to document your program's functionality and explain which hashing function you have chosen and why it's best for your program.

Time Complexity (4 points):

Mathematically determine the worst-case time complexity of your program for an input size of n . Remember to calculate the total amount of time your program takes, and provide a clear justification for your answer (Add a screenshot of your code that focuses only on the worst-case scenario, excluding constant-time parts). Submit this in a PDF document.

Example:

List of words: "apple", "banana", "orange", "grape", "peach"

User input: applle

Output: apple

Submission:

Your submission should include:

- The source code (with appropriate comments explaining the code logic).

- A brief report explaining:
 - Examples of how to use it.
 - Testing process and results.
 - Time complexity results in a PDF document.

Submit your C++ code files and the report as a single compressed file (ZIP file) through the Canvas platform.
